

Cloud Architecture

1. Pick any one app you use daily (like Zomato, BookMyShow, or Spotify) and write a short description of a new feature you would want to build for it that would require cloud hosting and scaling.

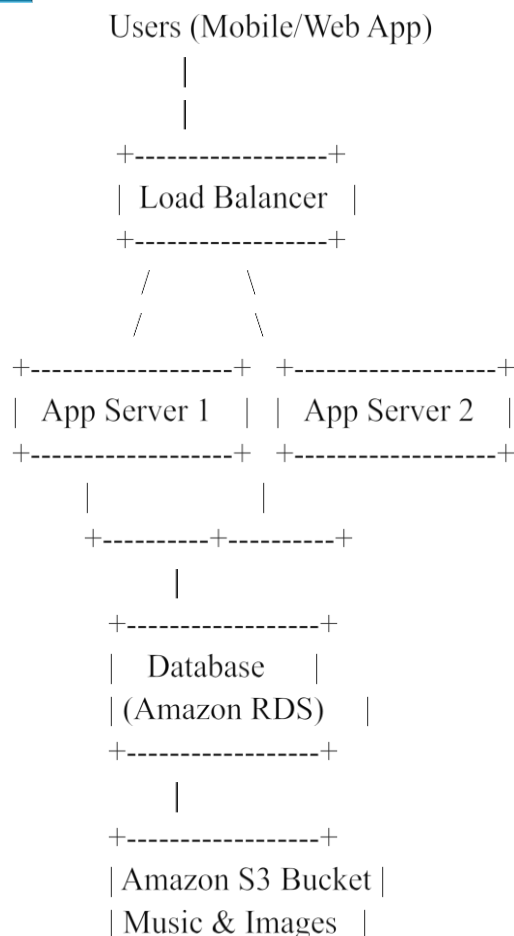
ANS :

- **Application:** Spotify
- **New Feature:** AI Mood-Based Playlist Generator

➤ I would like to add an **AI Mood-Based Playlist Generator** to Spotify. Users can select or describe their current mood (such as Happy, Sad, Relaxed, Workout, or Party), and the system will automatically generate a personalized playlist using AI. Since thousands or even millions of users may use this feature simultaneously, cloud hosting is required to provide scalable computing power, secure storage, and high availability.

2. Draw a basic cloud architecture diagram (hand-drawn or using a tool like draw.io) for your chosen feature, including at least: a load balancer, two app servers, a database, and a storage service (like AWS S3 or Azure Blob). Label each component clearly.

ANS :



❖ **Components:**

- **Load Balancer:** Distributes user requests across multiple servers.
- **App Servers:** Process user requests and generate AI playlists.
- **Database (Amazon RDS):** Stores user accounts, playlists, and preferences.

3. List three security measures you would add to your architecture to protect user data and prevent attacks. For each, explain in 1-2 lines how it helps (e.g., HTTPS, firewalls, authentication methods).

ANS :

Security Measure	Explanation
HTTPS (SSL/TLS)	Encrypts communication between users and the application to protect sensitive information from attackers.
AWS Security Groups / Firewalls	Allow only authorized traffic to reach the application and database while blocking unauthorized access.
Multi-Factor Authentication (MFA)	Adds an extra layer of security by requiring a second verification step before users can access their accounts.

4. Write a short document (200-300 words) explaining how your architecture would handle a sudden spike in users during a big event (like IPL finals for BookMyShow or a major album release on Spotify). Describe which parts would scale and how.

ANS :

- The proposed cloud architecture is designed to handle large numbers of users efficiently during major events, such as the release of a popular album on Spotify. When many users access the application at the same time, the **Load Balancer** distributes incoming requests evenly across multiple application servers. This prevents any single server from becoming overloaded and improves response time.
- The application servers are placed in an **Auto Scaling Group**. If CPU usage or incoming traffic increases beyond a predefined threshold, AWS automatically launches additional EC2 instances. As traffic decreases, the extra instances are terminated, which helps reduce operational costs.
- The **Amazon RDS** database stores user accounts, playlists, and listening history. It can also be configured with read replicas and automatic backups to improve performance and ensure data availability. **Amazon S3** stores music files, album artwork, and other static content. Since S3 is highly scalable, it can serve millions of requests without affecting application performance.

- Security is maintained through HTTPS encryption, Security Groups, and Multi-Factor Authentication, ensuring that user data remains protected. Overall, this architecture provides high availability, scalability, reliability, and security, allowing Spotify to continue serving users smoothly even during sudden spikes in demand.
5. Present your architecture design to a friend or family member, ask them for one piece of feedback or a question, and write down what they said along with your response.

Hint: Focus on explaining your choices simply, as if you were talking to someone new to cloud computing.

ANS :

Person: My friend

Feedback/Question: "Why do you need two application servers instead of one?"

My Response: Two application servers improve reliability and performance. If one server fails, the other server continues handling user requests, ensuring that the application remains available. Multiple servers also help distribute traffic efficiently during periods of high demand.